1

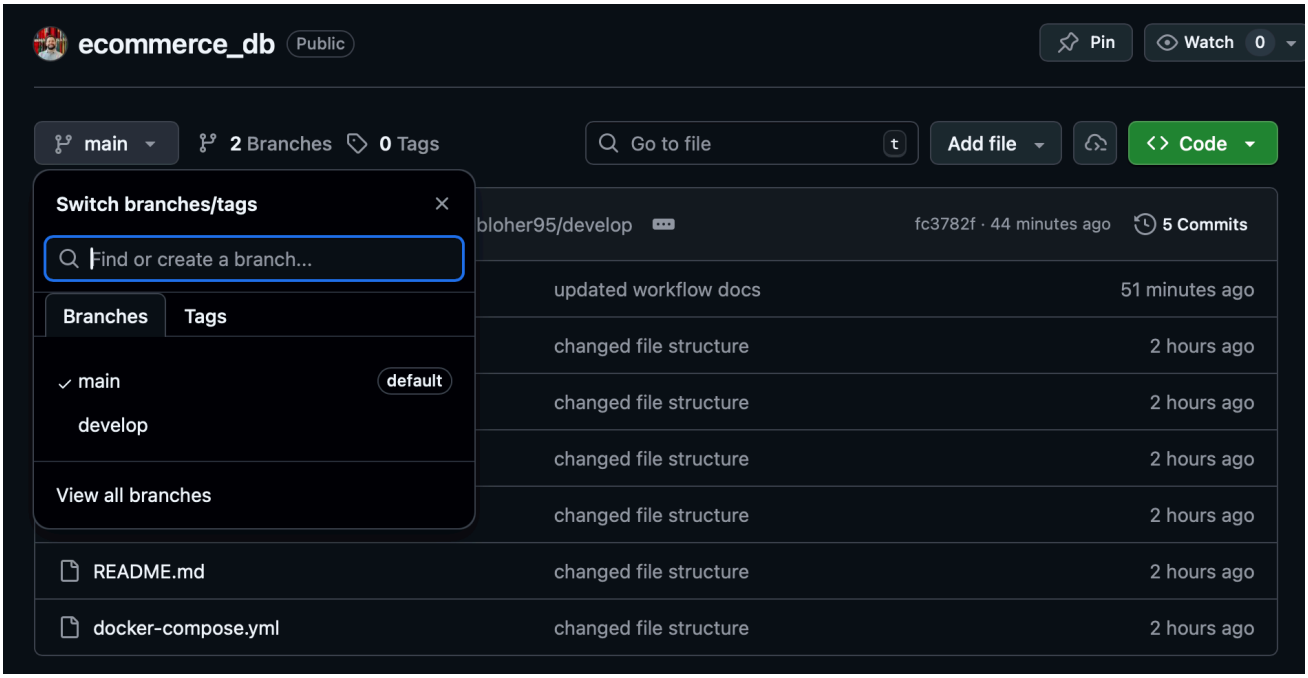
Git Repos:

https://github.com/pabloher95/ecommerce_db
https://github.com/pabloher95/ecommerce_products
https://github.com/pabloher95/ecommerce_orders
https://github.com/pabloher95/ecommerce_frontend

Git workflow docs: https://github.com/pabloher95/ecommerce_db/blob/main/docs/git_flow.md

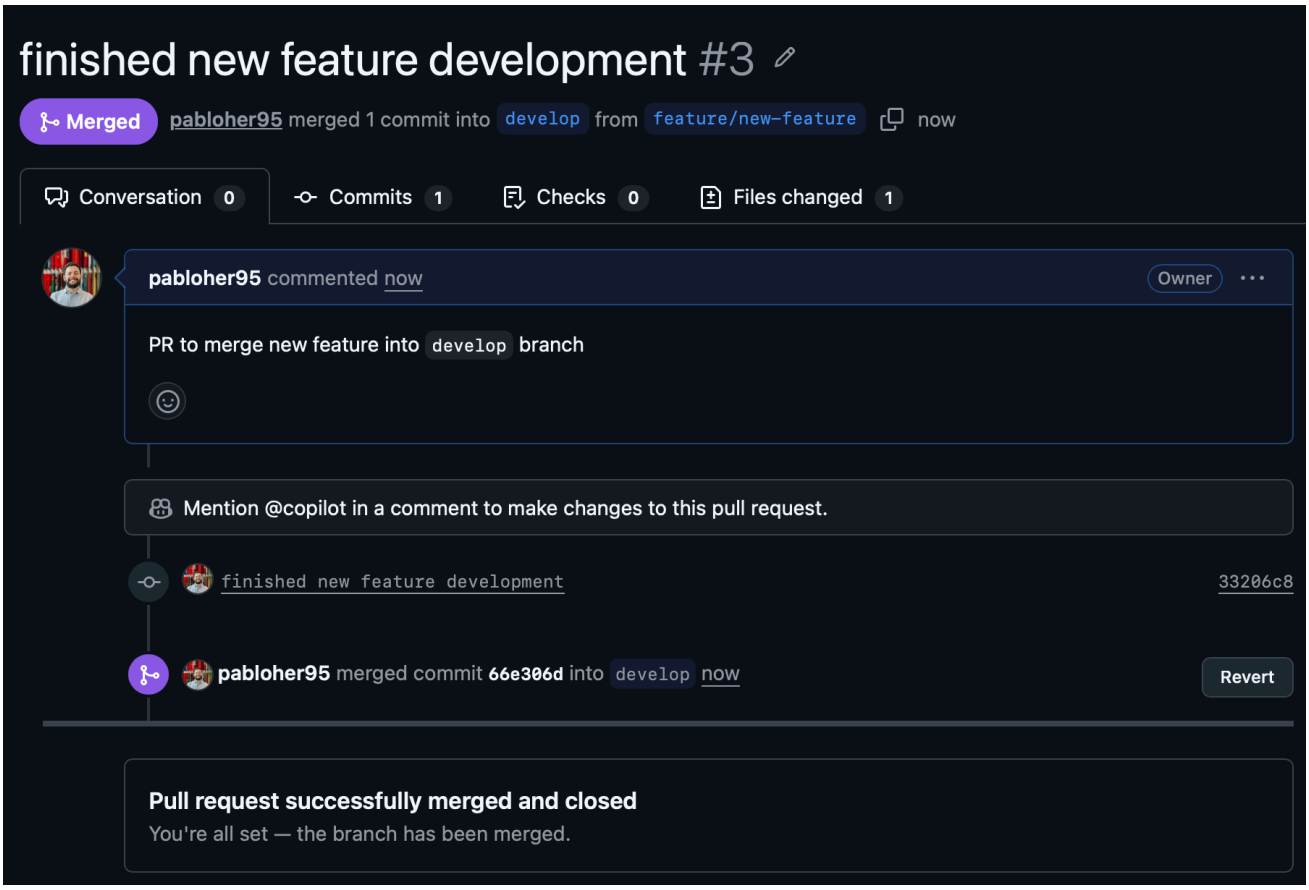
Screenshots:

- Initial setup of main and develop



- complete feature development workflow

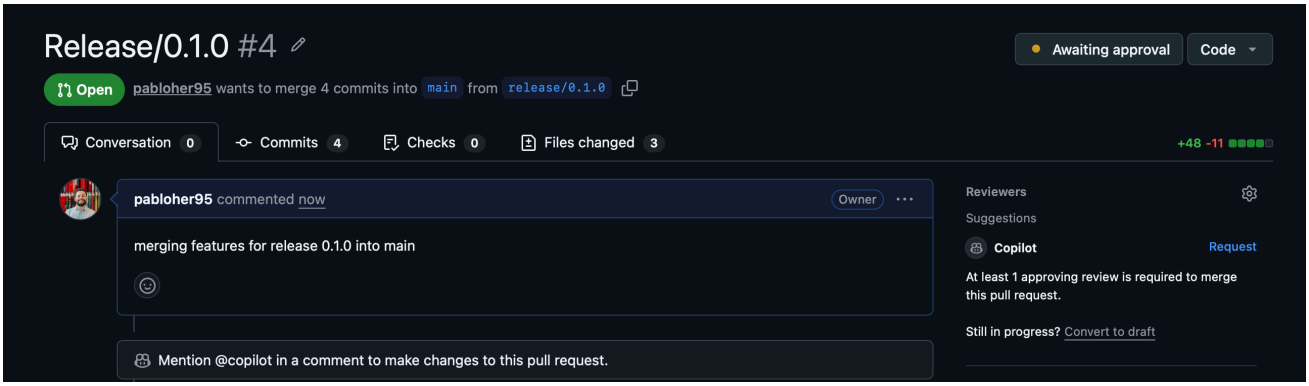
```
● pablohernandez@Pablos-MacBook-Pro-2 database % git branch
  * develop
    main
● pablohernandez@Pablos-MacBook-Pro-2 database % git checkout -b feature/new-feature
  Switched to a new branch 'feature/new-feature'
● pablohernandez@Pablos-MacBook-Pro-2 database % git add -u
● pablohernandez@Pablos-MacBook-Pro-2 database % git commit -m 'finished new feature development'
  [feature/new-feature 33206c8] finished new feature development
   1 file changed, 7 insertions(+)
● pablohernandez@Pablos-MacBook-Pro-2 database % git push --set-upstream origin feature/new-feature
  Enumerating objects: 5, done.
  Counting objects: 100% (5/5), done.
  Delta compression using up to 8 threads
  Compressing objects: 100% (3/3), done.
  Writing objects: 100% (3/3), 779 bytes | 779.00 KiB/s, done.
  Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
  remote: Resolving deltas: 100% (1/1), completed with 1 local object.
  remote:
  remote: Create a pull request for 'feature/new-feature' on GitHub by visiting:
  remote:      https://github.com/pabloher95/ecommerce_db/pull/new/feature/new-feature
  remote:
  To github.com:pabloher95/ecommerce_db.git
   * [new branch]      feature/new-feature -> feature/new-feature
  branch 'feature/new-feature' set up to track 'origin/feature/new-feature'.
```

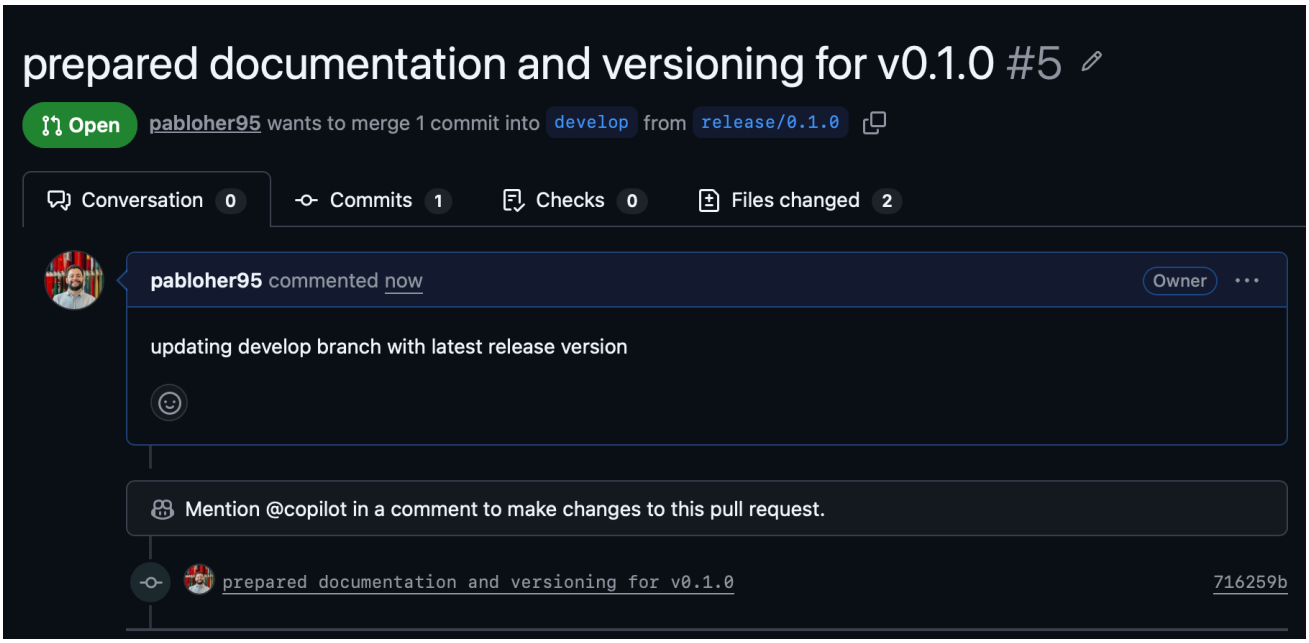


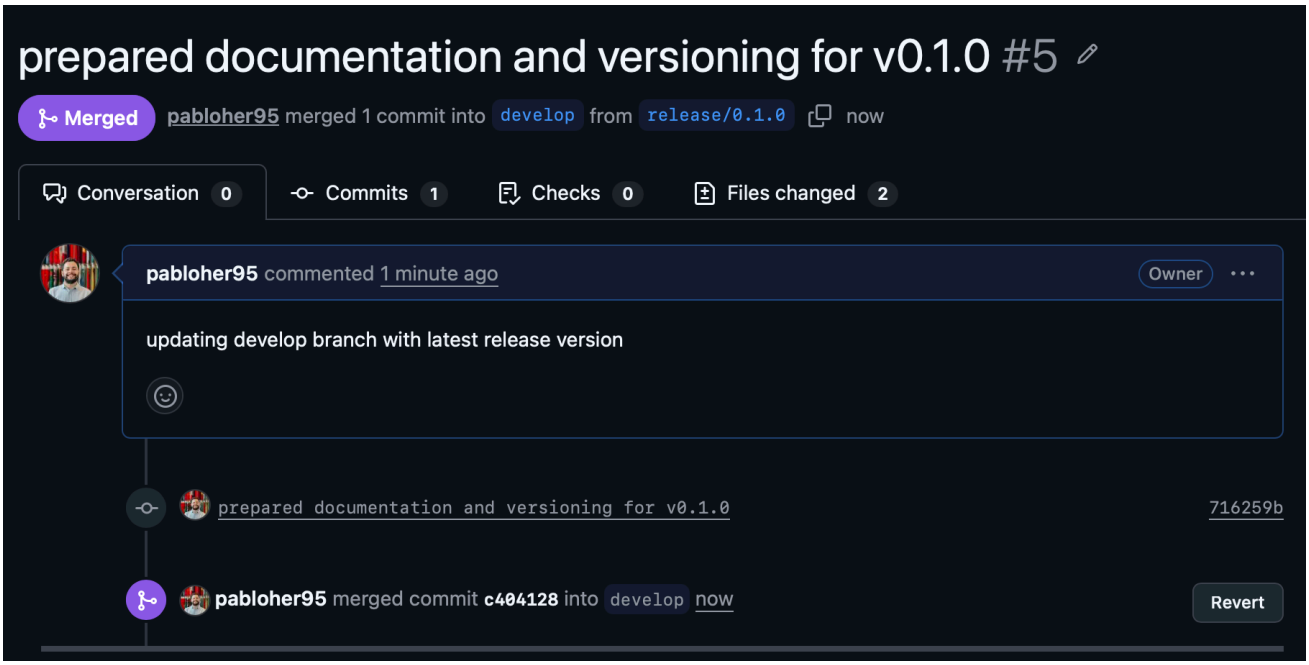

```
pablohernandez@Pablos-MacBook-Pro-2 database % git log --oneline -5
66e306d (HEAD -> develop, origin/develop) Merge pull request #3 from pabloher95/feature/new-feature
33206c8 (feature/new-feature) finished new feature development
bf4ad82 updated readme and env vars
80f1776 updated workflow docs
1406b68 updated workflow docs
pablohernandez@Pablos-MacBook-Pro-2 database % git branch -d feature/new-feature
Deleted branch feature/new-feature (was 33206c8).
```

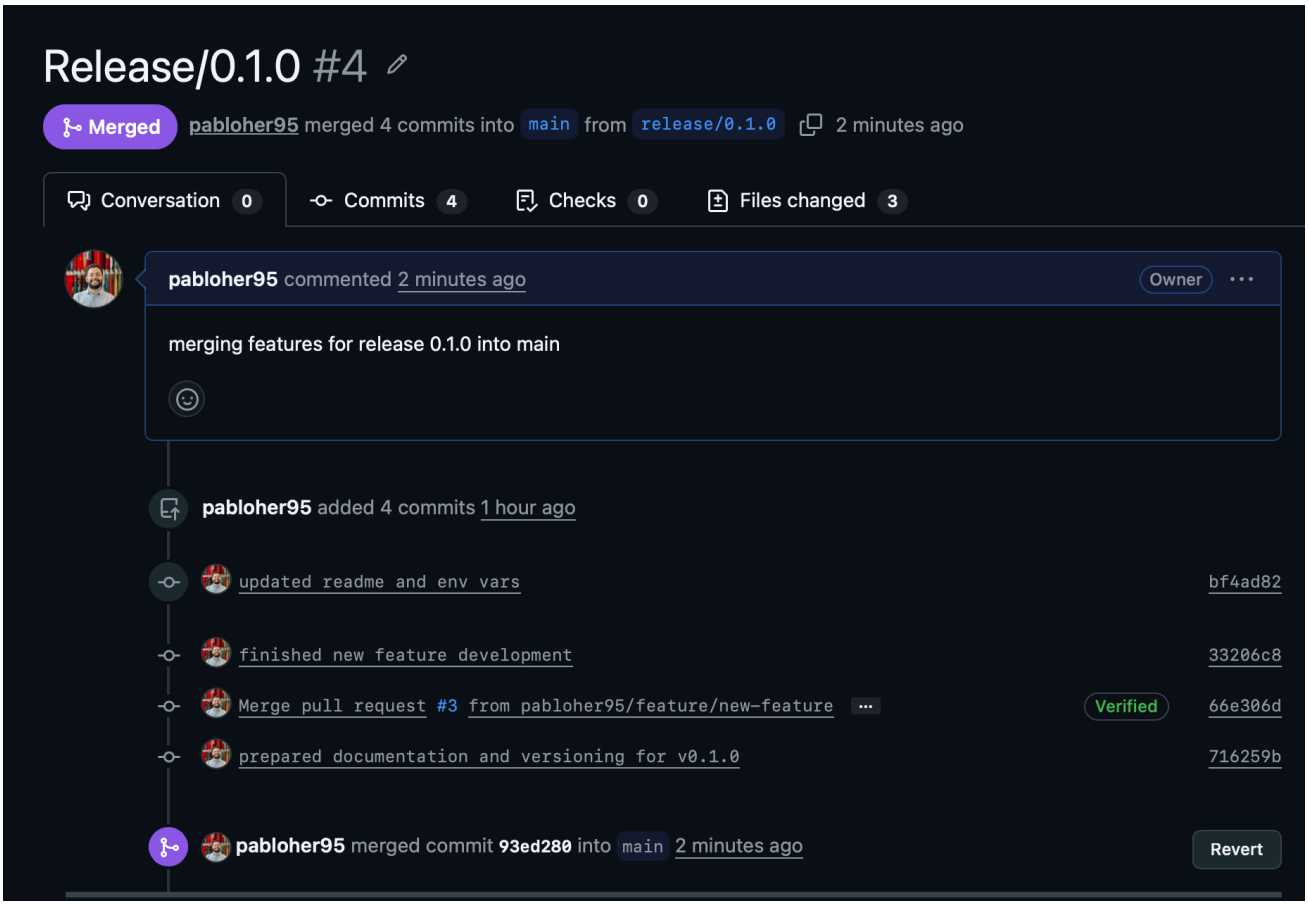
- complete release workflow

```
pablohernandez@Pablos-MacBook-Pro-2 database % git checkout -b release/0.1.0
Switched to a new branch 'release/0.1.0'
pablohernandez@Pablos-MacBook-Pro-2 database % git tag -a v0.1.0 -m "Release 0.1.0"
pablohernandez@Pablos-MacBook-Pro-2 database % git add -u
pablohernandez@Pablos-MacBook-Pro-2 database % git commit -m 'prepared documentation and versioning for v0.1.0'
[release/0.1.0 716259b] prepared documentation and versioning for v0.1.0
 2 files changed, 36 insertions(+), 12 deletions(-)
pablohernandez@Pablos-MacBook-Pro-2 database % git push -d origin release/0.1.0
error: unable to delete 'release/0.1.0': remote ref does not exist
error: failed to push some refs to 'github.com:pabloher95/ecommerce_db.git'
pablohernandez@Pablos-MacBook-Pro-2 database % git push --set-upstream origin release/0.1.0
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 1.30 KiB | 1.30 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'release/0.1.0' on GitHub by visiting:
remote:      https://github.com/pabloher95/ecommerce_db/pull/new/release/0.1.0
remote:
To github.com:pabloher95/ecommerce_db.git
 * [new branch]      release/0.1.0 -> release/0.1.0
branch 'release/0.1.0' set up to track 'origin/release/0.1.0'.
```



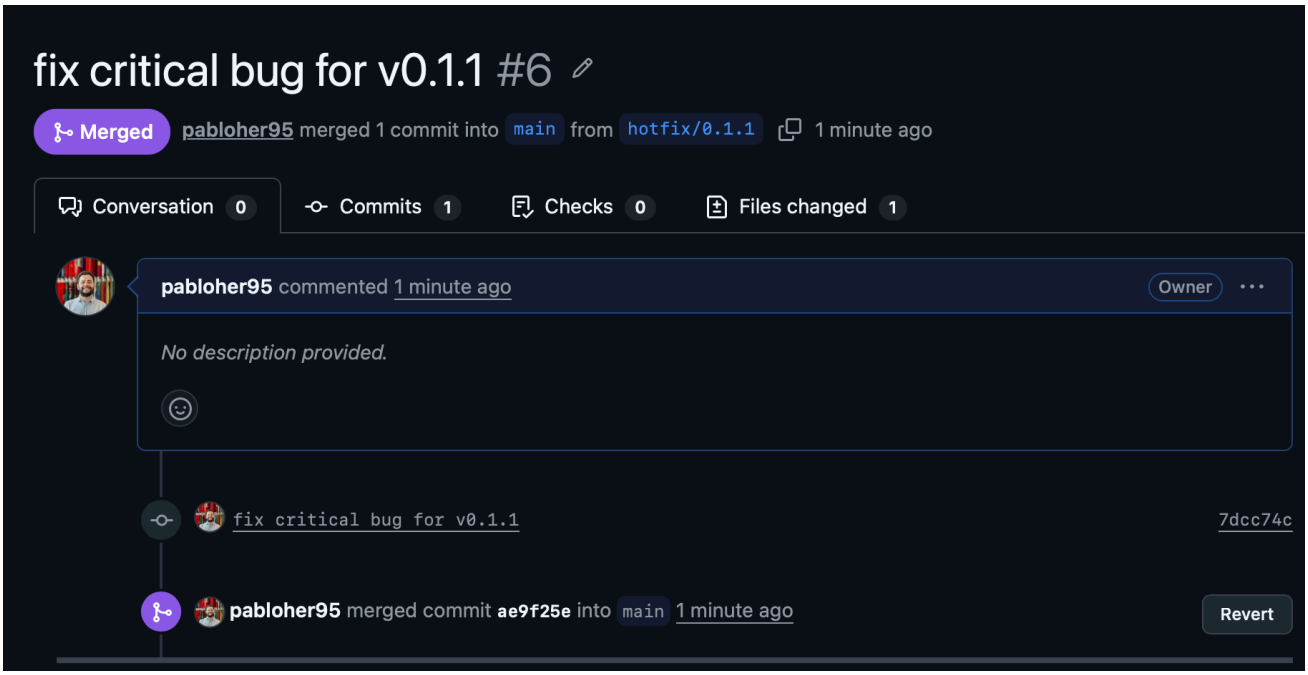



prepared documentation and versioning for v0.1.0 #5
Merged
pabloher95 merged 1 commit into develop from release/0.1.0 now
Conversation 0
Commits 1
Checks 0
Files changed 2
pabloher95 commented 1 minute ago
Owner
updating develop branch with latest release version
prepared documentation and versioning for v0.1.0
716259b
pabloher95 merged commit c404128 into develop now
Revert


Release/0.1.0 #4
Merged
pabloher95 merged 4 commits into main from release/0.1.0 2 minutes ago
Conversation 0
Commits 4
Checks 0
Files changed 3
pabloher95 commented 2 minutes ago
Owner
merging features for release 0.1.0 into main
pabloher95 added 4 commits 1 hour ago
updated readme and env vars
bf4ad82
finished new feature development
33206c8
Merge pull request #3 from pabloher95/feature/new-feature
Verified
66e306d
prepared documentation and versioning for v0.1.0
716259b
pabloher95 merged commit 93ed280 into main 2 minutes ago
Revert

```
pablohernandez@Pablos-MacBook-Pro-2 database % git checkout main
Switched to branch 'main'
Your branch is behind 'origin/main' by 3 commits, and can be fast-forwarded.
  (use "git pull" to update your local branch)
pablohernandez@Pablos-MacBook-Pro-2 database % git pull
remote: Enumerating objects: 2, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (2/2), 1.77 KiB | 454.00 KiB/s, done.
From github.com:pabloher95/ecommerce_db
   fc3782f..93ed280  main       -> origin/main
   66e306d..c404128  develop    -> origin/develop
Updating 9553e7b..93ed280
Fast-forward
 .env.example       |  5 +++++
 README.md          | 37 +++++++++++++++++++++++++-----------
 docker-compose.yml | 17 +++++++++++++++++
 docs/git_flow.md   | 75 ++++++++++++++++++++++++
```

fix critical bug for v0.1.1 #6
Open
pabloher95 wants to merge 1 commit into main from hotfix/0.1.1
Conversation 0
Commits 1
Checks 0
Files changed 1
pabloher95 commented now
Owner
No description provided.
Mention @copilot in a comment to make changes to this pull request.
fix critical bug for v0.1.1
7dcc74c


fix critical bug for v0.1.1 #6
Merged
pabloher95 merged 1 commit into main from hotfix/0.1.1
1 minute ago
Conversation 0
Commits 1
Checks 0
Files changed 1
pabloher95 commented 1 minute ago
Owner
No description provided.
fix critical bug for v0.1.1
7dcc74c
pabloher95 merged commit ae9f25e into main 1 minute ago
Revert

Hotfix/0.1.1 #7

Open pabloher95 wants to merge 3 commits into develop from hotfix/0.1.1

Conversation 0 | Commits 3 | Checks 0 | Files changed 1

pabloher95 commented now — Owner

Bringing in critical bug fix into develop for future releases

Mention @copilot in a comment to make changes to this pull request.

pabloher95 added 3 commits 2 hours ago

- Merge pull request #1 from pabloher95/develop — Verified — fc3782f
- Merge pull request #4 from pabloher95/release/0.1.0 — Verified — 93ed280
- fix critical bug for v0.1.1 — 7dcc74c

Hotfix/0.1.1 #7

Merged pabloher95 merged 3 commits into develop from hotfix/0.1.1 now

Conversation 0 | Commits 3 | Checks 0 | Files changed 1

pabloher95 commented now — Owner

Bringing in critical bug fix into develop for future releases

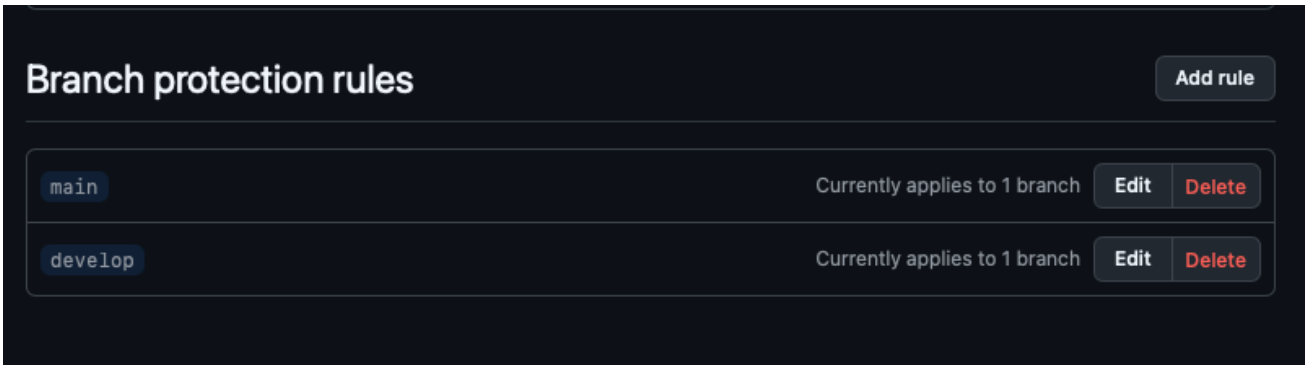
pabloher95 added 3 commits 2 hours ago

- Merge pull request #1 from pabloher95/develop — Verified — fc3782f
- Merge pull request #4 from pabloher95/release/0.1.0 — Verified — 93ed280
- fix critical bug for v0.1.1 — 7dcc74c

pabloher95 merged commit df774e3 into develop now — Revert

```
pablohernandez@Pablos-MacBook-Pro-2 database % git pull
remote: Enumerating objects: 2, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (2/2), 1.76 KiB | 300.00 KiB/s, done.
From github.com:pabloher95/ecommerce_db
   c404128..df774e3  develop    -> origin/develop
   93ed280..ae9f25e  main       -> origin/main
Already up to date.
pablohernandez@Pablos-MacBook-Pro-2 database % git checkout main
Switched to branch 'main'
Your branch is behind 'origin/main' by 2 commits, and can be fast-forwarded.
  (use "git pull" to update your local branch)
pablohernandez@Pablos-MacBook-Pro-2 database % git tag -a v0.1.1 -m "Hotfix 0.1.1"
pablohernandez@Pablos-MacBook-Pro-2 database % git push origin v0.1.1
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 171 bytes | 171.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:pabloher95/ecommerce_db.git
 * [new tag]         v0.1.1 -> v0.1.1
pablohernandez@Pablos-MacBook-Pro-2 database % git push -d origin hotfix/0.1.1
To github.com:pabloher95/ecommerce_db.git
 - [deleted]         hotfix/0.1.1
pablohernandez@Pablos-MacBook-Pro-2 database % git branch -d hotfix/0.1.1
error: The branch 'hotfix/0.1.1' is not fully merged.
If you are sure you want to delete it, run 'git branch -D hotfix/0.1.1'.
pablohernandez@Pablos-MacBook-Pro-2 database % git branch -D hotfix/0.1.1
Deleted branch hotfix/0.1.1 (was 7dcc74c).
```

- branch protection



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Dockerfiles:

https://github.com/pabloher95/ecommerce_db/blob/main/Dockerfile
https://github.com/pabloher95/ecommerce_products/blob/main/Dockerfile
https://github.com/pabloher95/ecommerce_orders/blob/main/Dockerfile
https://github.com/pabloher95/ecommerce_frontend/blob/main/Dockerfile

Name	Last Pushed ↑	Contains	Visibility	Scout
pabloher95/ecommerce-frontend	1 minute ago	IMAGE	Public	Inactive
pabloher95/ecommerce-order-service	1 minute ago	IMAGE	Public	Inactive
pabloher95/ecommerce-product-service	2 minutes ago	IMAGE	Public	Inactive
pabloher95/ecommerce-database	3 minutes ago	IMAGE	Public	Inactive

Docker hub repos:
https://hub.docker.com/repository/docker/pabloher95/ecommerce-database/general
https://hub.docker.com/repository/docker/pabloher95/ecommerce-product-service/general
https://hub.docker.com/repository/docker/pabloher95/ecommerce-order-service/general
https://hub.docker.com/repository/docker/pabloher95/ecommerce-frontend/general

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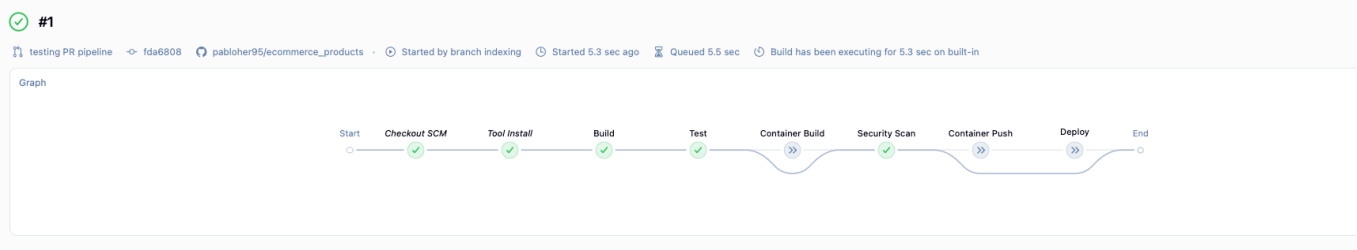
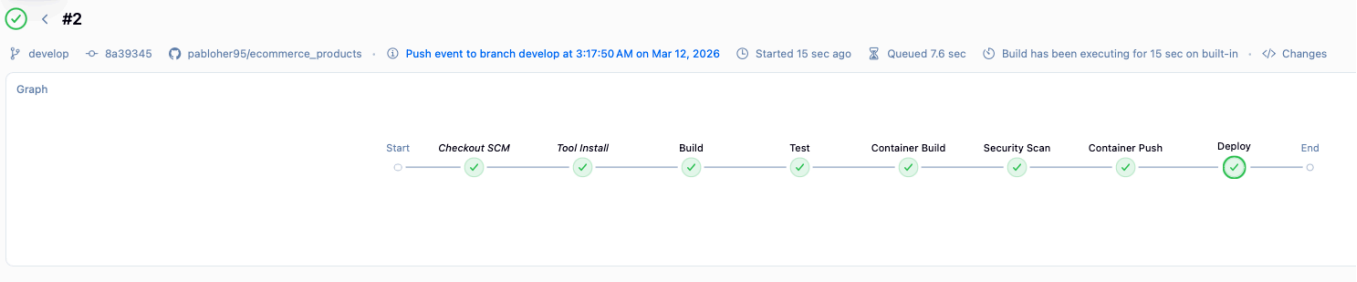
S	W	Name	Last Success ↑	Last Failure	Last Duration
		database	1 min 43 sec log	N/A	1.3 sec
		ecommerce-frontend	12 min log	N/A	1.2 sec
		order-service	14 min log	N/A	1 sec
		product-service	1 hr 0 min log	N/A	1 sec

Jenkinsfiles:
https://github.com/pabloher95/ecommerce_db/blob/main/Jenkinsfile
https://github.com/pabloher95/ecommerce_frontend/blob/main/Jenkinsfile
https://github.com/pabloher95/ecommerce_products/blob/main/Jenkinsfile
https://github.com/pabloher95/ecommerce_orders

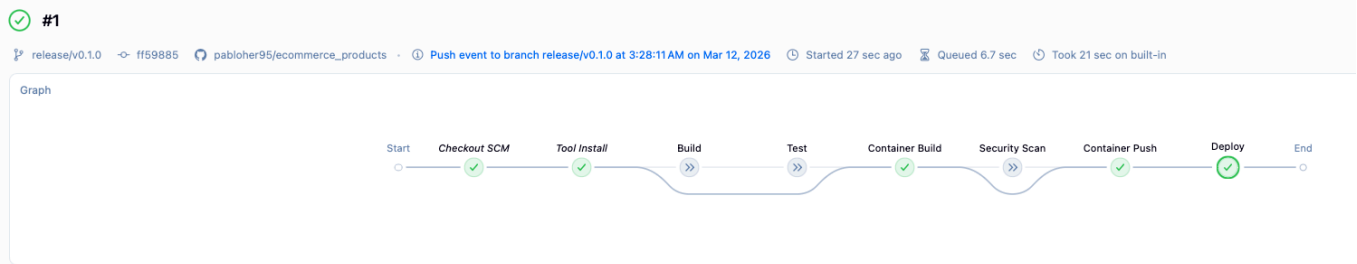
Build environment pipeline:



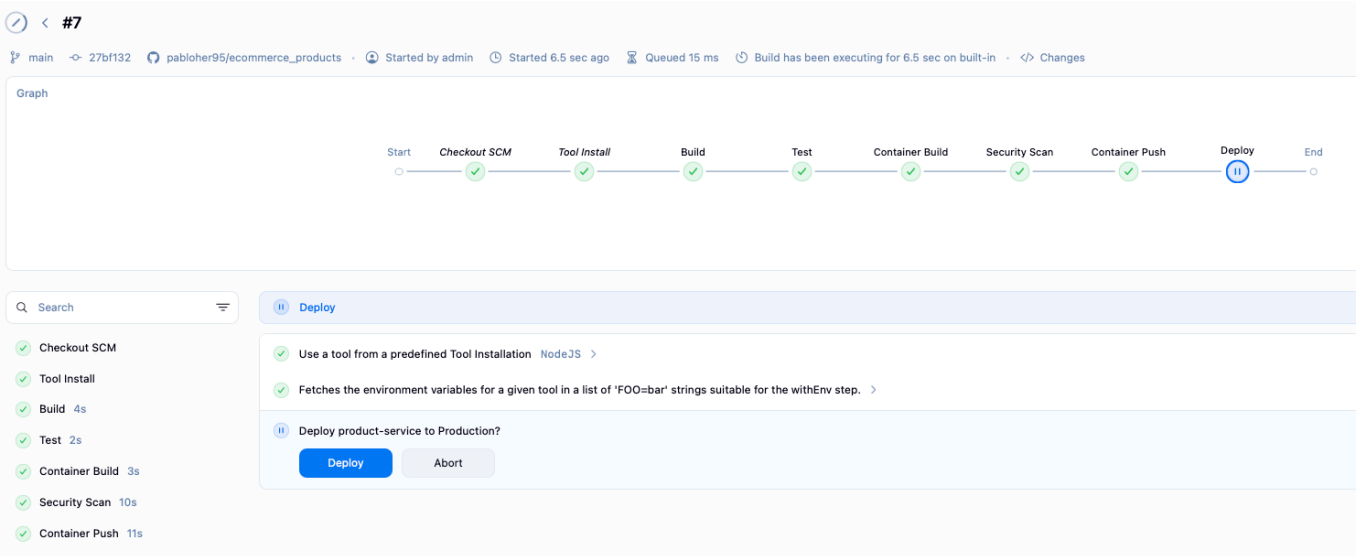
Dev environment pipeline:



Staging environment pipeline



Production environment pipeline



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dev:

```
pablohernandez@Pablos-MacBook-Pro-2 terraform % terraform plan -var-
file=env/dev.tfvars

Terraform used the selected providers to generate the following execution
plan. Resource actions
are indicated with the following symbols:
  + create
 <= read (data resources)

Terraform will perform the following actions:

  # module.jenkins.null_resource.jenkins will be created
  + resource "null_resource" "jenkins" {
      + id       = (known after apply)
      + triggers = {
          + "http_port" = "8080"
        }
    }
```

```
  # module.minikube.data.external.minikube_ip will be read during apply
  # (depends on a resource or a module with changes pending)
 <= data "external" "minikube_ip" {
      + id      = (known after apply)
      + program = [
          + "bash",
          + "-lc",
          + "minikube ip -p 'ecommerce-dev' | jq -R '{ip: .}'",
        ]
      + result  = (known after apply)
    }

  # module.minikube.null_resource.minikube will be created
  + resource "null_resource" "minikube" {
      + id       = (known after apply)
      + triggers = {
          + "addons"    = "ingress"
          + "cpus"      = "4"
          + "driver"    = "docker"
          + "memory_mb" = "6144"
          + "profile"   = "ecommerce-dev"
        }
    }

  # module.postgres.null_resource.postgres will be created
  + resource "null_resource" "postgres" {
      + id       = (known after apply)
      + triggers = {
          + "container_name" = "ecommerce-db-dev"
          + "db_name"        = "ecommerce"
          + "db_user"        = "postgres"
          + "host_port"      = "5432"
          + "image"          = "postgres:16"
          + "volume_name"    = "ecommerce-db-dev"
        }
    }

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + docker_registry     = "pabloher95"
  + jenkins_url         = "http://localhost:8080"
  + kube_context        = "minikube"
  + kubeconfig_path     = "/Users/pablohernandez/.kube/config"
  + postgres_connection = "postgresql://postgres@localhost:5432/ecommerce"
```

```
pablohernandez@Pablos-MacBook-Pro-2 terraform % terraform output
docker_registry = "pabloher95"
jenkins_url = "http://localhost:8080"
kube_context = "minikube"
kubeconfig_path = "/Users/pablohernandez/.kube/config"
postgres_connection = "postgresql://postgres@localhost:5432/ecommerce"
```

staging

```
pablohernandez@Pablos-MacBook-Pro-2 terraform % terraform plan -var-
file=env/staging.tfvars

Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with
the following symbols:
  + create
 <= read (data resources)

Terraform will perform the following actions:

  # module.jenkins.null_resource.jenkins will be created
  + resource "null_resource" "jenkins" {
      + id       = (known after apply)
      + triggers = {
          + "http_port" = "8081"
        }
    }

  # module.minikube.data.external.minikube_ip will be read during apply
  # (depends on a resource or a module with changes pending)
 <= data "external" "minikube_ip" {
      + id      = (known after apply)
      + program = [
          + "bash",
          + "-lc",
          + "minikube ip -p 'ecommerce-staging' | jq -R '{ip: .}'",
        ]
      + result  = (known after apply)
    }

  # module.minikube.null_resource.minikube will be created
  + resource "null_resource" "minikube" {
      + id       = (known after apply)
      + triggers = {
          + "addons"    = "ingress"
          + "cpus"      = "4"
```

```
          + "driver"    = "docker"
          + "memory_mb" = "6144"
          + "profile"   = "ecommerce-staging"
        }
    }

  # module.postgres.null_resource.postgres will be created
  + resource "null_resource" "postgres" {
      + id       = (known after apply)
      + triggers = {
          + "container_name" = "ecommerce-db-staging"
          + "db_name"        = "ecommerce"
          + "db_user"        = "postgres"
          + "host_port"      = "5433"
          + "image"          = "postgres:16"
          + "volume_name"    = "ecommerce-db-staging"
        }
    }

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + docker_registry     = "pabloher95"
  + jenkins_url         = "http://localhost:8081"
  + kube_context        = "minikube"
  + kubeconfig_path     = "/Users/pablohernandez/.kube/config"
  + postgres_connection = "postgresql://postgres@localhost:5433/ecommerce"
  +
```

```
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

docker_registry = "pabloher95"
jenkins_url = "http://localhost:8081"
kube_context = "minikube"
kubeconfig_path = "/Users/pablohernandez/.kube/config"
postgres_connection = "postgresql://postgres@localhost:5433/ecommerce"
```

prod

```
pablohernandez@Pablos-MacBook-Pro-2 terraform % terraform plan -var-
file=env/prod.tfvars

Terraform used the selected providers to generate the following execution
plan. Resource actions are indicated with
```

```
the following symbols:
  + create
 <= read (data resources)

Terraform will perform the following actions:

  # module.jenkins.null_resource.jenkins will be created
  + resource "null_resource" "jenkins" {
      + id       = (known after apply)
      + triggers = {
          + "http_port" = "8082"
        }
    }

  # module.minikube.data.external.minikube_ip will be read during apply
  # (depends on a resource or a module with changes pending)
 <= data "external" "minikube_ip" {
      + id      = (known after apply)
      + program = [
          + "bash",
          + "-lc",
          + "minikube ip -p 'ecommerce-prod' | jq -R '{ip: .}'",
        ]
      + result  = (known after apply)
    }

  # module.minikube.null_resource.minikube will be created
  + resource "null_resource" "minikube" {
      + id       = (known after apply)
      + triggers = {
          + "addons"    = "ingress"
          + "cpus"      = "4"
          + "driver"    = "docker"
          + "memory_mb" = "6144"
          + "profile"   = "ecommerce-prod"
        }
    }

  # module.postgres.null_resource.postgres will be created
  + resource "null_resource" "postgres" {
      + id       = (known after apply)
      + triggers = {
          + "container_name" = "ecommerce-db-prod"
          + "db_name"        = "ecommerce"
          + "db_user"        = "postgres"
          + "host_port"      = "5434"
```

```
          + "image"          = "postgres:16"
          + "volume_name"    = "ecommerce-db-prod"
        }
    }

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + docker_registry     = "pabloher95"
  + jenkins_url         = "http://localhost:8082"
  + kube_context        = "minikube"
  + kubeconfig_path     = "/Users/pablohernandez/.kube/config"
  + postgres_connection = "postgresql://postgres@localhost:5434/ecommerce"
```

```
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

docker_registry = "pabloher95"
jenkins_url = "http://localhost:8082"
kube_context = "minikube"
kubeconfig_path = "/Users/pablohernandez/.kube/config"
postgres_connection = "postgresql://postgres@localhost:5434/ecommerce"
```

Running Minikube cluster

```
pablohernandez@Pablos-MacBook-Pro-2 terraform % minikube status -p ecommerce-dev
ecommerce-dev
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

pablohernandez@Pablos-MacBook-Pro-2 terraform % kubectl get nodes
NAME             STATUS   ROLES           AGE   VERSION
ecommerce-prod   Ready    control-plane   14m   v1.35.0
pablohernandez@Pablos-MacBook-Pro-2 terraform % kubectl config current-context
ecommerce-prod
pablohernandez@Pablos-MacBook-Pro-2 terraform % kubectl config current-context
pablohernandez@Pablos-MacBook-Pro-2 terraform % terraform output minikube_ip
"192.168.49.2"
```

output for Jenkins

```
pablohernandez@Pablos-MacBook-Pro-2 terraform % terraform output -json
{
  "docker_registry": {
    "sensitive": false,
    "type": "string",
    "value": "pabloher95"
  },
  "jenkins_url": {
    "sensitive": false,
    "type": "string",
    "value": "http://localhost:8080"
  },
  "kube_context": {
    "sensitive": false,
    "type": "string",
    "value": "minikube"
  },
  "kubeconfig_path": {
    "sensitive": false,
    "type": "string",
    "value": "/Users/pablohernandez/.kube/config"
  },
  "minikube_ip": {
    "sensitive": false,
    "type": "string",
    "value": "192.168.49.2"
  },
  "postgres_connection": {
    "sensitive": false,
    "type": "string",
    "value": "postgresql://postgres@localhost:5432/ecommerce"
  }
}
```

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Manifests

https://github.com/pabloher95/ecommerce_orchestration/tree/main/k8s

Pods

```
pablohernandez@Pablos-MacBook-Pro-2 k8s % kubectl get pods -n dev
NAME                                      READY   STATUS    RESTARTS   AGE
ecommerce-database-86457ddfcb-8n85k       1/1     Running   0          2m43s
ecommerce-frontend-7884dfb469-9j47f       1/1     Running   0          2m43s
ecommerce-order-service-d7c4b85dd-4dlcn   1/1     Running   0          2m43s
ecommerce-product-service-898556f48-l282g 1/1     Running   0          2m43s
```

```
pablohernandez@Pablos-MacBook-Pro-2 k8s % kubectl get pods -n prod
NAME                                        READY   STATUS    RESTARTS   AGE
ecommerce-database-86457ddfcb-sljlt         1/1     Running   0          87s
ecommerce-frontend-7884dfb469-8zlbn         1/1     Running   0          87s
ecommerce-order-service-d7c4b85dd-2xr4q     1/1     Running   0          87s
ecommerce-product-service-898556f48-867q9   1/1     Running   0          87s
```

```
pablohernandez@Pablos-MacBook-Pro-2 k8s % kubectl get pods -n staging
NAME                                        READY   STATUS    RESTARTS   AGE
ecommerce-database-86457ddfcb-vljs8         1/1     Running   0          7s
ecommerce-frontend-7884dfb469-fgqtc         1/1     Running   0          7s
ecommerce-order-service-d7c4b85dd-lcc88     1/1     Running   0          7s
ecommerce-product-service-898556f48-7tvd8   1/1     Running   0          7s
```

Services

```
pablohernandez@Pablos-MacBook-Pro-2 k8s % kubectl get svc -n dev
NAME                                TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
ecommerce-database                  ClusterIP   10.98.233.171    <none>        5432/TCP         21s
ecommerce-frontend                  ClusterIP   10.106.254.72    <none>        80/TCP           21s
ecommerce-frontend-nodeport         NodePort    10.106.130.106   <none>        80:30000/TCP     21s
ecommerce-order-service             ClusterIP   10.111.124.80    <none>        3002/TCP         21s
ecommerce-order-service-nodeport    NodePort    10.110.216.187   <none>        3002:30002/TCP   21s
ecommerce-product-service           ClusterIP   10.108.178.231   <none>        3001/TCP         21s
ecommerce-product-service-nodeport  NodePort    10.100.63.85     <none>        3001:30001/TCP   21s
```

Deployment rollout status

```
pablohernandez@Pablos-MacBook-Pro-2 k8s % kubectl rollout status deployment -n dev
deployment "ecommerce-database" successfully rolled out
deployment "ecommerce-frontend" successfully rolled out
deployment "ecommerce-order-service" successfully rolled out
deployment "ecommerce-product-service" successfully rolled out
```